

PREDICTING EMPLOYEE ATTRITION USING MACHINE LEARNING

Machine Learning Model Comparison for HR Decision Support



Loren Martin

Python | Scikit-learn | Random Forest | AdaBoost | Power BI

Portfolio Case Study

Executive Summary

Employee attrition can create recruiting, onboarding, training, productivity, and knowledge-retention costs for an organization (Ruehl, 2024). This project evaluates whether historical workforce characteristics can help identify employees who may be at greater risk of leaving and compares predictive approaches that support different HR priorities.

The analysis was completed in two stages. An initial logistic-regression baseline model used a small feature set to establish a benchmark. The baseline achieved 66.7% testing accuracy, 40.3% testing precision, and 29.1% testing recall. A later workflow expanded the feature set, incorporated exploratory analysis and one-hot encoding, and compared Random Forest and AdaBoost ensemble classifiers. Both ensemble models achieved approximately 85% testing accuracy. Random Forest produced higher recall (59%), while AdaBoost produced higher precision (92%).

Primary takeaway

The strongest model depends on the business objective. Random Forest is more useful when HR wants to identify a broader share of employees who may leave; AdaBoost is more useful when HR wants fewer false positives and more selective intervention.

Business Problem and Objective

The primary analytical question was: Can historical workforce characteristics help identify employees who may be at greater risk of attrition? The analysis also examined how model choice changes when HR values broader identification of attrition risk versus more targeted predictions.

Data Sources and Preparation

The analysis used two course-provided human-resources datasets containing employee and organizational characteristics. The datasets were merged using the EmployeeNumber field to create a unified analytical dataset. The original source files are not redistributed in this portfolio repository because no explicit redistribution license was provided.

- Project One established a baseline model using a derived HighIncome indicator and YearsWithCurrManager.
- Project Two expanded the modeling data with workforce characteristics including age, job satisfaction, monthly income, total working years, years at company, and related organizational variables.
- Categorical variables used in the final encoded dataset included gender, marital status, job role, and department.

Baseline Model

A logistic regression model was used as the baseline benchmark. The data were split into 80% training and 20% testing sets using a fixed random state. The baseline intentionally used a limited feature set, providing a reference point for evaluating whether later feature engineering and ensemble methods improved predictive performance.

Metric	Training	Testing	Interpretation
Accuracy	72.7%	66.7%	Overall correct classifications
Precision	49.2%	40.3%	Reliability of positive attrition predictions
Recall	36.9%	29.1%	Share of employees who left that the model identified

The baseline model showed similar training and testing accuracy, but its low testing precision and recall indicated limited practical ability to identify employees who ultimately left. This result established a clear benchmark for the expanded modeling phase.

Exploratory Data Analysis

The expanded analysis first examined the workforce outcome distribution and selected employee characteristics. Approximately 27.4% of employees experienced attrition, while 72.6% remained. This class imbalance is important because accuracy alone can overstate performance when the majority outcome dominates the data (Owusu-Adjei et al., 2023).

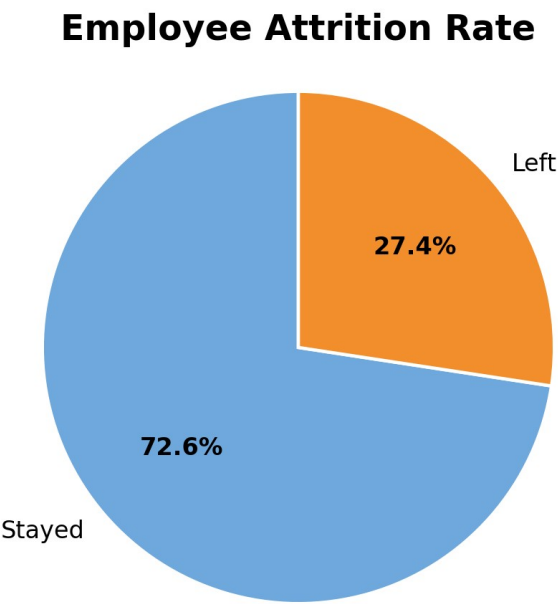


Figure 1. Employee attrition outcome distribution.

Monthly income and years at the company were also examined by attrition status. Employees who left showed greater variability in both measures, but the distributions overlapped substantially. The exploratory findings therefore supported evaluating multiple workforce characteristics together rather than relying on a single variable to explain turnover.

Feature Engineering and Model Development

The expanded workflow created additional compensation and tenure indicators and selected a broader group of workforce features for supervised classification. A `high_income` indicator was set when `MonthlyIncome` was at least 7,000, and a `Tenured` indicator was set when `YearsAtCompany` was at least 11. Gender, marital status, job role, and department were converted to numeric indicator columns using one-hot encoding.

The final model data were split into 70% training and 30% testing sets. Two ensemble classification approaches were evaluated: Random Forest and AdaBoost. Grid search was used in the original notebook workflow to tune model parameters before performance was assessed with accuracy, precision, and recall.

Model Performance

Model	Testing Accuracy	Testing Precision	Testing Recall
Baseline Logistic Regression	66.7%	40.3%	29.1%
Random Forest	85%	84%	59%
AdaBoost	85%	92%	55%

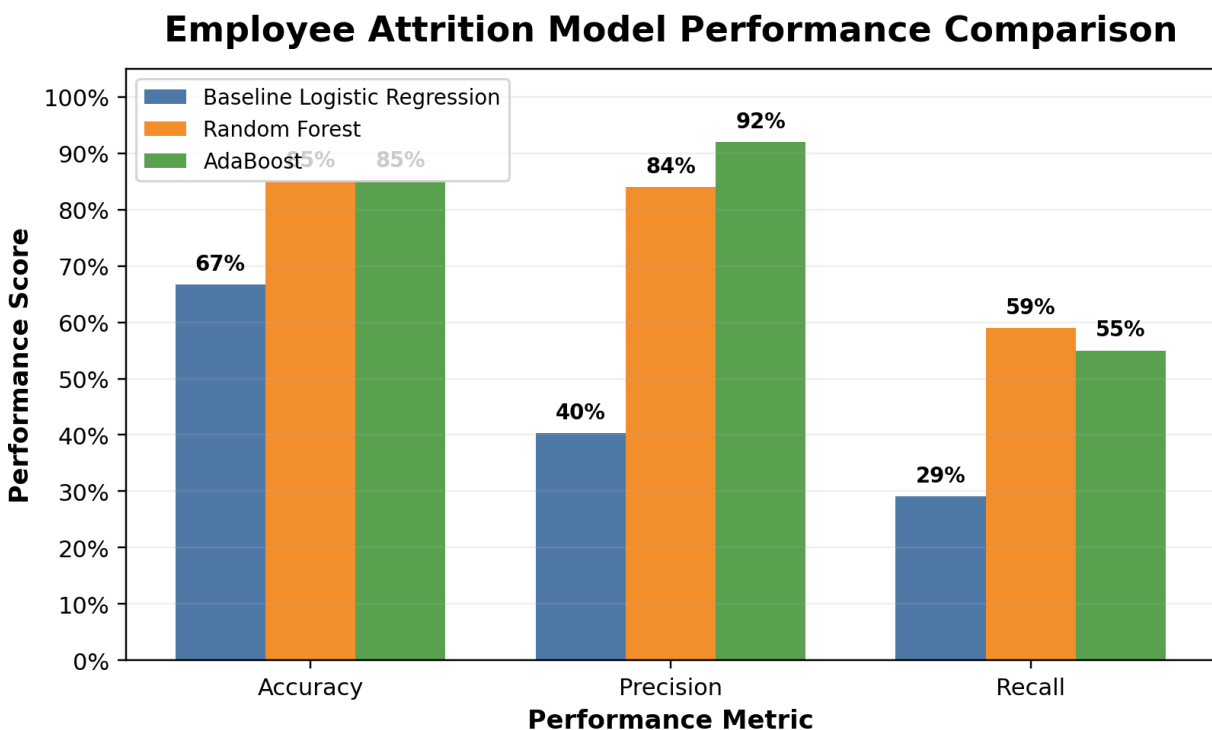


Figure 2. Testing performance across the baseline, Random Forest, and AdaBoost models.

Both ensemble models improved substantially on the baseline test metrics. Random Forest achieved 59% recall, identifying a larger share of employees who ultimately left. AdaBoost achieved 92% precision, producing more reliable positive predictions and fewer false positives. Although both models reached approximately 85% testing accuracy, their precision-recall tradeoff leads to different business uses (Scikit-learn developers, n.d.).

Business Interpretation

- Random Forest is preferable when the priority is broader identification of attrition risk and HR wants to miss fewer potentially at-risk employees.
- AdaBoost is preferable when intervention resources are limited and HR wants positive predictions to be more selective and reliable.
- Accuracy should not be used as the sole model-selection criterion; precision and recall provide context that directly affects resource allocation and intervention strategy.
- Predictions should be treated as decision support and considered alongside broader workforce information rather than as standalone employment decisions.

Recommendations

- Use model outputs to prioritize proactive retention review, not to make automatic personnel decisions.
- Select Random Forest when reaching more employees who may be at risk is the primary goal.
- Select AdaBoost when reducing false positives and targeting scarce retention resources is the primary goal.
- Monitor performance as workforce composition and attrition patterns change, and retrain models when updated employee data become available.
- Continue evaluating multiple workforce characteristics together rather than relying on a single factor such as income or tenure.

Limitations and Portfolio Considerations

- The attrition outcome is imbalanced: 27.4% left and 72.6% stayed. This makes precision and recall particularly important alongside accuracy.
- The baseline and ensemble stages used different feature sets and different train/test split configurations, so the performance comparison demonstrates analytical progression rather than a strictly controlled head-to-head experiment.
- Random Forest showed much stronger training than testing performance in the original notebook (99% training accuracy versus 85% testing accuracy), which suggests that generalization should continue to be monitored on new data.
- Portfolio consideration: because attrition models can affect people, any real-world operational use would require fairness review, privacy safeguards, human oversight, and evaluation of whether model features create inappropriate or discriminatory effects.

Data Availability

The original HRData1.csv and HRData2.csv files were provided through university coursework and are not redistributed in this portfolio repository. Public artifacts focus on the analytical workflow, summarized results, visualizations, model evaluation, and business interpretation.

Portfolio Project Background

This portfolio case study consolidates two applied data analytics coursework projects into a single end-to-end narrative. The first stage established a logistic-regression benchmark, while the second stage expanded exploratory analysis, feature engineering, ensemble modeling, and business-focused model comparison.

Reproducibility Note

The public portfolio version preserves the analytical workflow and summarized results without redistributing course-provided row-level data. The cleaned Jupyter Notebook documents the original merge, preprocessing, encoding, training, and evaluation steps and provides portfolio-safe outputs for review.

References

Owusu-Adjei, M., Hayfron-Acquah, J. B., Frimpong, T., & Abdul-Salaam, G. (2023). Imbalanced class distribution and performance evaluation metrics: A systematic review of prediction accuracy for determining model performance in healthcare systems. *PLoS Digital Health*, 2(11). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10688675/>

Ruehl, E. (2024, April 15). Attrition: Definition, types, causes & mitigation tips. SHRM. <https://www.shrm.org/topics-tools/news/employee-relations/attrition-definition-types-causes-mitigation-tips>

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